

Lab-4: CS-2203 Artificial Intelligence Lab

Assignment No: 04

Date: Jan 28, 2026

Submission Deadline: [Jan 28, 2026 - 17:00 Hrs. IST]

Instructions for Submission

General Instructions

Some assignments may need to be submitted via Google Form. Please carefully follow the instructions below to ensure proper submission:

File Preparation

Submission Instructions:

1. Store all assignment-related files in a single folder and compress it into a `.rar` or `.zip` file. The compressed file must be named in the following format:

`roll-number_assign4.rar` or `roll-number_assign4.zip`

2. For example, if your roll number is 2201AI01, the compressed file should be named as:

`2201AI01_assign4.rar` or `2201AI01_assign4.zip`

3. Save each program using the filename format specified alongside each question.
4. If all assignments are completed within a single Jupyter Notebook, name the notebook file using the following format:

`CS2203_Lab4.ipynb`

5. When using a single notebook for multiple problems, clearly **segregate the code for each problem**. Each problem must be introduced using a **text cell** containing the problem statement before the corresponding code segment.
6. After completing the notebook, place it inside the assignment folder, compress the folder, and upload it by following the above naming instructions.

7. Additionally, submit a **PDF containing both the code and its output**. To generate the PDF:

- Open the Jupyter Notebook file.
- Navigate to **File** → **Print**.
- Save the file as a PDF using the same name as the notebook (e.g., `CS2204_Lab3.pdf`).

Submission Link

Upload your compressed file using the following Google Form link:

<https://forms.gle/iEWvirKXX2MM33Gt9>

Problem Statement 1 (20 marks + 10 Viva)

You are required to design and implement the Convolutional Neural Network (CNN) architecture depicted in **Figure 1** using **TensorFlow/Keras Functional API**.

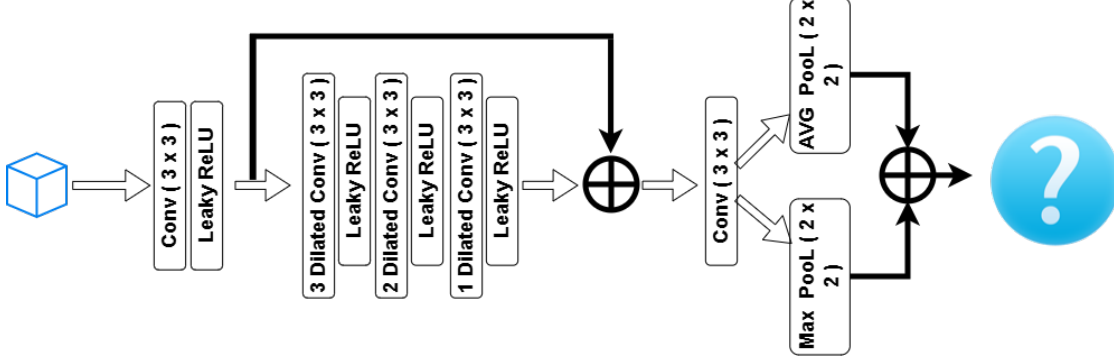


Figure 1: Architecture Diagram: Residual connections with Dilated Convolutions and Parallel Pooling.

Architecture Specifications

The network processes a fixed 5×5 single-channel input through the following stages:

1. **Initial Block:** A Conv2D layer (3×3) followed by a Leaky ReLU activation.
2. **Residual Path (Skip Connection):**
 - The output of the Initial Block is split into two paths.
 - **Path A (Skip):** Carries the data directly to the adder.
 - **Path B (Main):** Passes through three sequential **Dilated Convolutions** with dilation rates of $r = 3$, $r = 2$, and $r = 1$ respectively. Each convolution is followed by a Leaky ReLU.
 - **Merge:** Element-wise addition ($\oplus$) of Path A and Path B.
3. **Intermediate Layer:** A Conv2D layer (3×3) processing the merged output.
4. **Parallel Pooling Block:**
 - The output splits into two parallel pooling layers:
 - **Path 1:** Average Pooling (2×2).
 - **Path 2:** Max Pooling (2×2).
 - **Final Output:** Element-wise addition ($\oplus$) of the Average Pooling and Max Pooling outputs.

Note: Use `padding='same'` for all Convolution layers to maintain spatial dimensions for the add operations. Initialize all kernels to **ones** (`kernel_initializer='ones'`) to ensure deterministic results.

Input Data

Do not use random initialization. Construct the input tensor manually using the specific pixel values given in the matrix below.

The input image I (Shape: $1 \times 5 \times 5 \times 1$) is defined as:

$$I = \begin{bmatrix} 1 & 2 & 3 & 0 & 1 \\ 0 & 1 & 2 & 3 & 0 \\ 3 & 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 0 & 0 \\ 0 & 1 & 0 & 1 & 2 \end{bmatrix}$$

Lab Tasks

1. **Architecture Implementation:** Write the TensorFlow/Keras code to build the model described above.
2. **Model Summary:** Print the model summary to verify the layer connections and shapes.
3. **Feature Map Calculation:** Pass the provided Input Data I through your model and predict the output.
4. **Question:** What is the final output feature map value? (Print the resulting array).

Evaluation Criteria : 1

The evaluation of this lab assignment will be based on the following criteria:

1. **Correctness of Implementation (40%):**
2. **Code Quality & Q&A (30%):**
 - The code should be well-structured, properly commented, and follow standard Python programming practices.
 - Students may be asked questions related to their implementation to assess their understanding.
3. **Error Handling (10%):**
 - The code gracefully handles potential errors, such as file not found or invalid image formats.
4. **Documentation (10%):**
 - The code is accompanied by appropriate documentation and comments that explain the steps taken in each part of the program.
5. **Timeliness of Submission (10%):**
 - The assignment is submitted on time, adhering to the submission deadline.

Evaluation Criteria: 2

The evaluation of this lab assignment will be based on the following criterion:

1. **Viva (10 Marks):**

- Questions will be asked related to the assignment to assess the student's understanding of the implemented concepts.